This equation is valid for orthogonal lattices only. For non-orthogonal lattices, such an expression may not be obtained easily; one may need to find $\hat{\mathbf{n}}$ by some other method and then use Eq. (1.8) to determine d. For a cubic lattice, a, b and c are equal and we get

$$d = \frac{a}{\left(h^2 + k^2 + l^2\right)^{1/2}} \tag{1.10}$$

It may also be noted that for a cubic lattice, the direction $[hkl]$ is perpendicular to the plane (hkl).

1.10 SIMPLE CRYSTAL STRUCTURES

We shall now describe some of the basic crystal structures which are either monoatomic or contain simple basis. These include *close-packed structures* like hexagonal close-packed or face-centred cubic structures and *loose-packed structures* like body-centred cubic or simple cubic structures. Besides these, the structures of diamond, zinc blende and sodium chloride are also described.

1.10.1. Close-Packed Structures

Close-packed structures are mostly found in monoatomic crystals having non-directional bonding, such as metallic bonding. In these structures, the coordination number of each atom is 12, i.e., each atom is surrounded by twelve similar and equal sized neighbours. Out of these twelve neighbours, six lie in one plane, three in an adjacent parallel plane above this plane and three in a similar plane below it. There are two types of close-packed structures :

(i) Hexagonal close-packed (*hcp*) structure

(ii) Face-centred cubic (*fcc*) structure

These structures are described as follows :

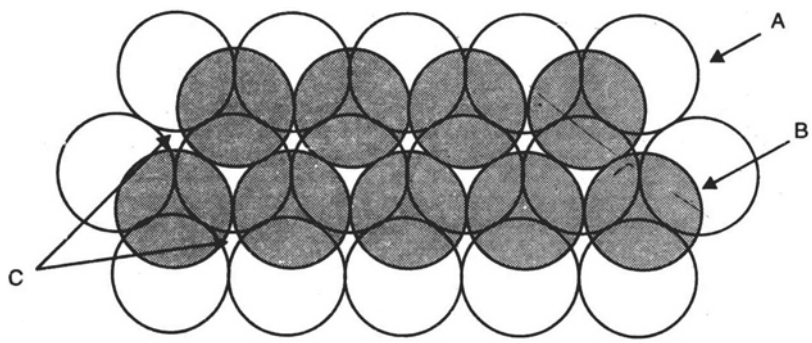


Fig. 1.18. *Layered arrangement of close-packed structures.*

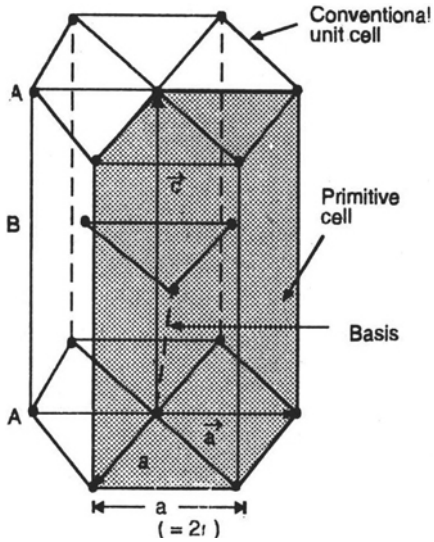


Fig. 1.19. *Conventional and primitive cells of hexagonal close-packed structure.*

(i) Hexagonal Close-Packed Structure

Consider a layer of similar atoms with each atom surrounded by six atoms in one plane as shown in Fig. 1.18. Another similar layer B can be placed on top of layer A such that the atoms of layer B occupy the alternate valleys formed by the atoms of layer A. If a third similar layer is placed on top of the B-layer in such a way that the atoms of B-layer exactly overlap the atoms of A-layer and this type of stacking is repeated successively, the following layered arrangement is obtained :

....ABABAB....

This type of stacking is called *hcp stacking* and the structure is known as *hexagonal close-packed structure.* The name corresponds to the shape of the conventional unit cell which is hexagonal and is shown in Fig. 1.19. There are twelve atoms located at the corners, two at the centres of the basal planes, and three completely inside the hexagon forming a part of the B-layer. The effective number of atoms in a unit cell is

$$12(1/6) + 2\ (1/2) + 3 = 6$$

The interatomic distance for the atoms within a layer is a. The distance between the two adjacent layers is $c/2$, c being the height of the unit cell. For an ideal *hcp* structure, $c = 1.633a$.

It may be noted that although the structure is *hcp*, the space lattice is simple hexagonal with basis consisting of two atoms placed in such a way that if one atom lies at the origin, the other atom lies at the point (2/3, 1/3, 1/2). The shaded portion in Fig. 1.19 represents the primitive cell of this structure. It contains 2 atoms instead of one which is due to the presence of the basis. Also, the volume of the primitive cell is exactly one-third of the volume of the hexagonal cell.

The *packing fraction, f,* is defined as the ratio of the volume occupied by the atoms present in a unit cell to the total volume of the unit cell. It is also referred to as the *packing factor* or *packing efficiency* of the unit cell. From the primitive cell, we find

$$f = \frac{2(4/3)\pi r^3}{a(a\sin 60^\circ)c}$$

where r is the atomic radius. Using $c = 1.633a$ and $a = 2r$, we get

$$f = 0.74$$

Thus, in an ideal *hcp* structure, 74% of the total volume is occupied by atoms. Metals like Mg, Zn, Cd, Ti, etc. exhibit this type of structure.

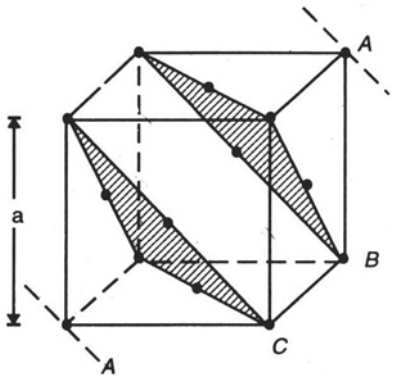


Fig. 1.20. *Conventional unit cell of fcc structure along with the stacking sequence .... ABCABC ....*

(ii) Face-Centred Cubic Structure

In this structure, the stacking of first two layers A and B is similar to that of hcp structure. The difference arises in the third layer which, in the present case, does not overlap the first layer. The atoms of the third layer occupy the positions of those valleys of the A-layer which are not occupied by the B-layer atoms. The third layer is designated by the letter C. The fourth layer exactly overlaps the first layer and the sequence is repeated. Thus *fcc* structure is represented by the following stacking sequence :

.... ABCABCABC ....

The conventional unit cell is *face-centred cubic* and is shown in Fig. 1.20. It is a non-primitive cell having effective number of atoms equal to 8 (1/8) + 6 (1/2) or 4. The atoms touch one another along the face diagonals. The length of the cube edge, a, and the atomic radius, r, are related to each other as

$$4r = \sqrt{2}a$$

The packing fraction, f, is given by

$$f = \frac{4(4/3)\pi r^3}{a^3} = 0.74$$

Thus the packing fraction of *fcc* structure is exactly the same as that of hcp structure which is expected because of the close-packed nature of both the structures. Also, the coordination number of each atom is 12. Examples of materials having this type of structure are Cu, Ag, Au, Al, etc.

1.10.2. Loose-Packed Structures

A loose-packed structure is that in which the coordination number of an atom is less than 12 or the packing fraction is less than 0.74. Among the various possible loose-packed structures, the most common and the simplest are the *body-centred cubic (bcc)* and the *simple cubic (sc)* structures. These structures are described as follows :

(i) Body-Centred Cubic Structure (bcc)

The conventional unit cell of *bcc* structure is non-primitive and is shown in Fig. 1.21. It has cubical shape with atoms located at the corners and the body centre. Thus the effective number of atoms per unit cell is 8 (1/8) + 1 = 2. The coordination number of each atom is 8. The atoms touch one another along the body diagonal. Thus a is related to r as

$$4r = \sqrt{3}\ a.$$

The packing fraction is given by

$$f = \frac{2(4/3)\pi r^3}{a^3} = 0.68$$

The examples of materials exhibiting *bcc* structure are Na, K, Mo, W, etc.

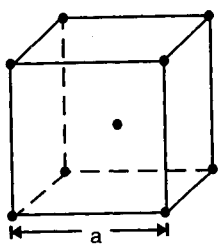


Fig. 1.21. *Conventional unit cell of bcc structure.*

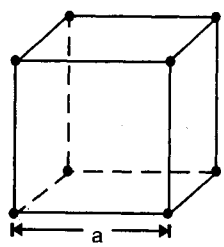


Fig. 1.22. *Unit cell of sc structure.*

(ii) Simple Cubic Structure (sc)

The conventional unit cell of *sc* structure is the same as its primitive cell and is shown in Fig. 1.22. The atoms are located at the corners only and touch one another along the cube edges. Thus in *sc* structures, we have

$$a = 2\ r$$

The coordination number of each atom is 6. The packing fraction is given by

$$f = \frac{1(4/3)\pi r^3}{a^3} = 0.52$$

Only polonium exhibits this type of structure at room temperature.

1.11 STRUCTURE OF DIAMOND

Diamond exhibits both cubic and hexagonal type structures. The diamond cubic (*dc*) structure is more common and is described here. The space lattice of the diamond cubic structure is *fcc* with basis consisting of two carbon atoms, one located at the lattice point and the other at a distance of one quarter of the body diagonal from the lattice point along the body diagonal. The unit cell of the *dc* structure is shown in Fig. 1.23. The carbon atoms placed along the body diagonals, in fact, occupy the alternate tetrahedral void positions in the *fcc* arrangement of carbon atoms. This opens up the otherwise close-packed *fcc* arrangement which decreases the packing efficiency considerably. The packing efficiency of the *dc* structure is only 34% as compared to 74% for the *fcc* structure. The coordination number of each carbon atom is 4 and the nearest neighbour distance is equal to $\sqrt{3}a/4$ where a is the lattice parameter.

The *dc* structure may also be viewed as an interpenetration of two *fcc* sublattices with their origins at (0, 0, 0) and (1/4, 1/4, 1/4). A plan view of the positions of all the carbon atoms in the unit cell is shown in Fig. 1.24. The fractional heights of the carbon atoms relative to the base of the unit cell are given in the circles drawn at the atomic positions. Two numbers in the same circle indicate two carbon atoms at the same position located one above the other. Other materials exhibiting this type of structure are Si, Ge, SiC, GaAs, gray tin, etc.

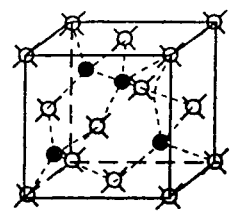


Fig. 1.23. *The unit cell of dc structure. The lattice is fcc with carbon atoms located at fcc positions and at alternate tetrahedral sites.*

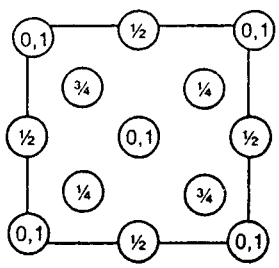


Fig. 1.24. *Plan view of atomic positions in dc unit cell. Numbers in the circles indicate fractional heights of the carbon atoms.*

1.12 ZINC BLENDE (ZnS) STRUCTURE

The zinc blende structure is similar to the *dc* structure except that the two *fcc* lattices in it are occupied by different elements. The structure is similar to the one shown in Fig. 1.23 where the dark circles now represent one type of atoms, say Zn, and the light circles represent the other type of atoms, i.e., S.

1.13 SODIUM CHLORIDE (NaCl) STRUCTURE

The unit cell of NaCl structure is shown in Fig. 1.25. In NaCl structure, the radii of Na^+ and Cl^- ions are such that each Na^+ ion is octahedrally coordinated to six Cl^- ions. The unit cell is *fcc* with four Cl^- ions occupying all the four *fcc* positions and the four Na^+ ions occupying all the four octahedral voids. The *fcc* positions and the octahedral void positions are, however, interchangeable. The NaCl structure can, therefore, be viewed as two interpenetrating *fcc* sublattices, one belonging to Na^+ ions with its origin at the point (0, 0, 0) and the other belonging to Cl^- ions with its origin at the point (a/2, 0, 0). In the erminology of lattice and basis, the structure can be interpreted as an *fcc* lattice with basis consisting of two ions, one of Na^+ and the other of Cl^-. One of these ions occupies one of the *fcc* positions and the other ion occupies the corresponding octahedral void position. A unit cell of NaCl comprises four molecules. The position of various ions in the unit cell are as follows :

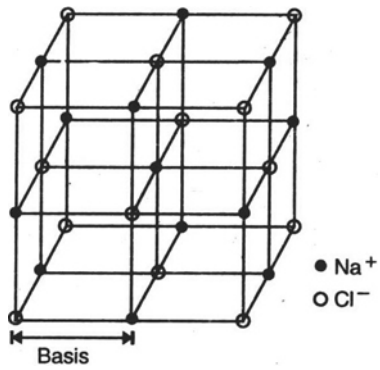


Fig. 1.25. *Unit cell of sodium chloride structure.*

Na^+ : 0, 0, 0; 1/2,1/2,0; 1/2,0,1/2; 0,1/2,1/2

Cl^- : 1/2,1/2,1/2; 0,0,1/2; 0,1/2,0; 1/2,0,0

SOLVED EXAMPLES

Example 1.1. Determine the relationships between the lattice parameter a and the atomic radius r for monoatomic *sc*, *bcc* and *fcc* structures.

Solution. In *sc* structure (Fig. 1.26), the atoms touch one another along the cube edges.

$$\therefore \quad a = 2r$$

In *bcc* structure, the atoms touch along the body diagonals.

$$\therefore \quad \sqrt{3}a = 4r$$

$$\text{or} \quad a = 4r/\sqrt{3}$$

In *fcc* structure, the atoms touch along the face diagonals.

$$\therefore \quad \sqrt{2}a = 4r$$

$$\text{or} \quad a = 2\sqrt{2}r$$

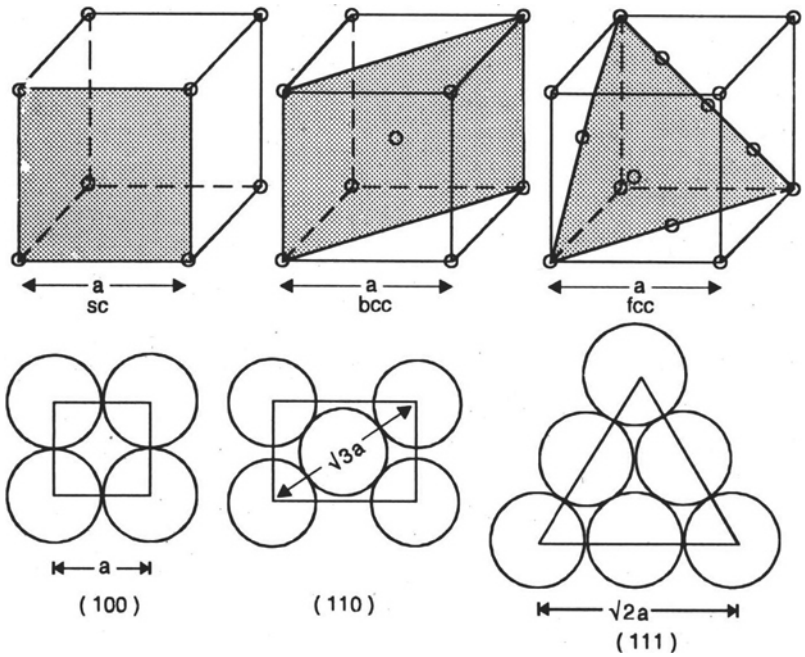


Fig. 1.26. *Monoatomic sc, bcc and fcc structures along with their (100), (110) and (111) type planes respectively.*

Example 1.2. Draw (101) and (111) planes in a cubic unit cell. Determine the Miller indices of the directions which are common to both the planes.

Solution. Intercepts of the plane (101) with the axes = 1/1, 1/0 and 1/1

= 1, ∞ and 1

Intercepts of the plane (111) with the axes = 1, 1 and 1

Taking the point O as origin and the lines OA, OB and OC as the axes **a**, **b** and **c** respectively, the plane with intercepts 1, ∞ and 1 is the plane ADGC and that with intercepts 1, 1 and 1 is plane ABC as shown in Fig. 1.27. Therefore, the line common to both the planes is the line AC. It corresponds to two directions, i.e., AC and CA.

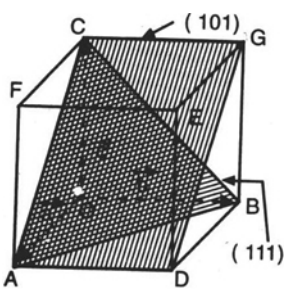


Fig. 1.27. *Planes (101) and (111) in a cubic lattice.*

Projections of the direction AC on the axes = −1, 0 and 1

Projections of the direction CA on the axes = 1, 0 and −1

Therefore, the required indices are $[\bar{1}01]$ and $[10\bar{1}]$.

Example 1.3. A plane makes intercepts of 1, 2 and 0.5Å on the crystallographic axes of an orthorhombic crystal with $a:b:c = 3:2:1$. Determine the Miller indices of this plane.

Solution. Taking the lengths of the axes OA, OB and OC as 3, 2 and 1Å respectively, the plane with intercepts of 1, 2 and 0.5Å on the axes is the plane DBE as shown in Fig. 1.28. The intercepts of this plane relative to full lengths of the axes are

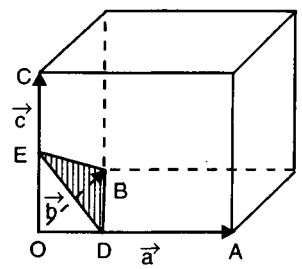


Fig. 1.28. *Plane (312) in an orthorhombic lattice.*

1/3, 2/2 and 0.5/1

or 1/3, 1 and 1/2

Reciprocals : 3, 1 and 2

Therefore, the Miller indices of the plane DBE are (312).

Example 1.4. In a cubic unit cell, find the angle between normals to the planes (111) and (121).

Solution. Since the crystal is cubic, the normals to the planes (111) and (121) are the directions [111] and [121] respectively. Let θ be the angle between the normals.

$$\therefore \quad \cos\theta = \frac{h_1h_2 + k_1k_2 + l_1l_2}{(h_1^2 + k_1^2 + l_1^2)^{1/2}\ (h_2^2 + k_2^2 + l_2^2)^{1/2}}$$

$$= \frac{1\times1 + 1\times2 + 1\times1}{(1^2 + 1^2 + 1^2)^{1/2}\ (1^2 + 2^2 + 1^2)^{1/2}}$$

$$= 0.9428$$

or $\theta = 19.47^{o}$ or $19^{o}28'$

Example 1.5. Calculate the packing efficiency and density of sodium chloride from the following data :

Radius of sodium ion = 0.98 Å

Radius of chloride ion = 1.81 Å

Atomic mass of sodium = 22.99 amu

Atomic mass of chlorine = 35.45 amu

Solution. The unit cell of NaCl strucutre is shown in Fig. 1.25. The Na^+ and Cl^- ions touch along the cube edges.

$\therefore$ Lattice parameter, a = 2 (Radius of Na^+ + Radius of Cl^-)

= 2 (0.98 + 1.81) = 5.58 Å

$$\text{Packing fraction} = \frac{\text{Volume of ions present in the unit cell}}{\text{Volume of the unit cell}}$$

$$= \frac{4(4/3)\pi r_{Na^+}^3 + 4(4/3)\pi r_{Cl^-}^3}{a^3}$$

$$= \frac{16\pi}{3}\left[\frac{(0.98)^3 + (1.81)^3}{(5.58)^3}\right]$$

$$= 0.663 \text{ or } 66.3\%$$

$$\text{Density} = \frac{\text{Mass of the unit cell}}{\text{Volume of the unit cell}}$$

$$= \frac{4(22.99 + 35.45) \times 1.66 \times 10^{-27}}{\left(5.58 \times 10^{-10}\right)^3} \text{ kg m}^{-3}$$

$$= 2234 \text{ kg m}^{-3} \text{ or } 2.23 \text{ g cm}^{-3}$$

SUMMARY

1. The solids may be broadly classified as crystalline and non-crystalline (or amorphous). The crystalline solids may be further sub-divided into single crystals and polycrystalline materials.

2. Crystallography is the study of formation, structure and properties of crystals.

3. A crystal structure results from the combination of a space lattice and a basis. A space lattice is a regular arrangement of infinite number of imaginary points in three-dimensional space. A basis is a structural unit comprising a single atom or a group of atoms which is placed on each lattice point in a regular fashion to generate the crystal structure.

4. A unit cell is a small group of points which acts as a building block for the entire lattice. It may be primitive or non-primitive. A primitive cell is the smallest volume unit cell and contains only one lattice point per cell. A non-primitive cell contains more than one lattice points per cell. The conventional unit cell has the highest possible symmetry and the lowest possible volume. It may be primitive or non-primitive.

5. The effective number of lattice points belonging to a unit cell is

$$N = N_i + N_f/2 + N_c/8$$

where N_i, N_f and N_c denote the number of lattice points present inside, at the face centres, and at the corners of the cell respectively.

6. A crystal remains invariant under the application of various symmetry operations like translation, rotation, reflection, inversion etc. Some rotational operations, such as 5-fold and 7-fold rotations, are not permissible as these are not compatible with lattice translation symmetry.

7. A point group is the combination of certain symmetry operations like rotation, reflection and inversion. It determines the symmetry of space around a point. The number of point groups in three-dimensional space is 32. These point groups produce only 14 Bravais lattices.

8. The set of all the symmetry elements of a crystal structure is called the space group. The number of distinct space groups possible in three dimensions is 230.

9. The Miller indices of a crystallographic plane and a direction are denoted by (*hkl*) and [*hkl*] respectively where *h*, *k* and *l* are integers. The parallel planes and the parallel directions have the same indices.

10. The angle between two directions [*hkl*] and [*h'k'l'*] is given by

$$\cos\theta = \frac{hh' + kk' + ll'}{(h^2 + k^2 + l^2)^{1/2}\,(h'^2 + k'^2 + l'^2)^{1/2}}$$

11. The interplanar distance for the parallel (*hkl*) planes for an orthorhombic lattice is

$$d = (h^2/a^2 + k^2/b^2 + l^2/c^2)^{-1/2}$$

where *a*, *b* and *c* are the lengths of the axes.

12. A close-packed structure is that in which each atom has twelve identical nearest neighbours. A close-packed structure may be either *fcc* or *hcp* with the following sequence of layers :

hcp : .....ABABABAB.....

fcc : .....ABCABCABC.....

VERY SHORT QUESTIONS

1. Define a single crystal.
2. How does a crystal differ from a grain?
3. What is short-range order?
4. What are amorphous materials? Give an example of such a material.
5. What is crystallography?
6. What is a unit cell?
7. How does a crystal differ from a lattice?
8. What is the maximum number of possible Bravais lattices?
9. What are Miller indices? What is their importance?
10. Write the indices of all the twelve edges of a cube.
11. Write the indices of all the face diagonals of a cube.
12. What is packing efficiency? What are its values for sc, *bcc*, *fcc* and *hcp* structures?
13. Give at least one example each of materials exhibiting *sc*, *bcc*, *fcc* and *hcp* structures.
14. Calculate the number of carbon atoms per unit cell of diamond.

SHORT QUESTIONS

1. Define primitive and non-primitive translation vectors. Which type of translation vectors are preferred for describing a lattice?
2. Prove that the crystals cannot have five-fold symmetry.
3. Differentiate primitive cell, non-primitive cell and conventional cell from one another. How is a Weigner-Seitz cell constructed?
4. Describe the scheme to determine the Miller indices of a plane. Show that the parallel planes have the same Miller indices.
5. What is the relationship between the Miller indices and reciprocal lattice vectors corresponding to any plane?
6. Find the Miller indices of cube faces and diagonal planes of a unit cube.

7. Explain the concepts of lattice, basis and crystal structure. How are they related ?
8. Draw primitive cells corresponding to *bcc* and *fcc* unit cells.
9. How does *hcp* structure differ from *bcc* structure?
10. What is Bravais lattice? What is the maximum number of Bravais lattices possible? How will you account for the existence of thousands of structures from these lattices?
11. The end-centred orthorhombic is one of the Bravais lattices but the end-centred tetragonal is not. Give reasons.
12. The primitive cell of *fcc* lattice is rhombohedral. Why then is the rhombohedral lattice included separately in the Bravais list?
13. State the points of similarity and difference of the monoatomic, *sc*, monoatomic *bcc*, and CsCl structures?
14. Calculate the volume of the primitive cell and the number of nearest neighbours for an *fcc* lattice.
15. Obtain an expression for the packing fraction for *hcp* structure.
16. Show that the *c/a* ratio for an ideal *hcp* lattice is $\sqrt{8/3}$.
17. Determine the values of packing fraction for *fcc*, *bcc* and *sc* structures.
18. Assuming one of the basis atoms lying at the origin, find the coordinates of the other atoms for an *hcp* structure.
19. Explain, without calculation, why *fcc* and *hcp* structures have the same packing factor.
20. Show that for a cubic lattice, the lattice constant, a, is given by

$$a = \sqrt[3]{\frac{n\,M}{N\,\rho}}$$

where the symbols have their usual meanings.

21. What type of lattice and basis do the following structures have :

(i) Sodium chloride (ii) Diamond cubic ?

22. Diamond is the hardest substance known in spite of the fact that the packing fraction and the coordination number of carbon atom in the *dc* structure are quite low. Explain.
23. There are four vacant tetrahedral sites in a unit cell of the *dc* structure. Can four additional carbon atoms occupy these sites? Give reasons.
24. How many crystal directions constitute the family of body diagonals of a unit cube ? Draw all such directions.

LONG QUESTIONS

1. What are symmetry operations? Describe the principal symmetry operations applicable to a three-dimensional lattice. Show that the five-fold rotational axis is not permissible in case of lattices.
2. What are point group and space group? Give their number for two-and three-dimensional lattices. List all the point groups of a two- dimensional lattice.
3. Determine the interplanar spacing between the two parallel planes with Miller indices (h, k, l) in a cubic crystal of side a.
4. Which is the most densely packed structure amongst the various cubic structures? Determine the packing fraction and porosity of this structure. Can the porosity be reduced by some means? What type of solids generally exhibit this type of structure and why?
5. Draw a plan view of sodium chloride structure. In how many ways can this structure be interpreted?
6. Draw a plan view of *hcp* unit cell and give coordinates of all the atoms. Are all the atoms located at equivalent sites? Discuss implications of your answer.
7. Draw the following:

   (*i*) $[1\bar{1}1]$, $[1\bar{2}1]$ and $[0\bar{1}2]$ directions in cubic and tetragonal lattices.

   (*ii*) $(\bar{1}11)$, $(1\bar{1}2)$ and $(2\bar{1}0)$ planes in cubic and orthorhombic lattices.

PROBLEMS

1. Find the Miller indices for planes with each of the following sets of intercepts:

   (*i*) $3a$, $3b$, $2c$; (*ii*) a, $2b$, ∞; (*iii*) $5a$, $-6b$, c;
   (*iv*) a, $b/2$, c; (*vi*) a, b,$-c$; (*vi*) $a/2$, b, ∞

   where a, b and c are lattice parameters.

   ((223), (210), $(6\bar{5}\,30)$, (121), (111), (210))

2. Draw a $(1\bar{1}0)$ plane in a cubic unit cell. Show all the <111> directions that lie on this plane and give the Miller indices of each direction.

   $([111], [11\bar{1}], [\bar{1}\,\bar{1}1]$ and $[\bar{1}\,\bar{1}\,\bar{1}])$

4. A plane makes intercepts of 1, 2 and 3 Å on the crystallographic axes of an orthorhombic crystal with $a : b : c = 3 : 2 : 1$. Determine the Miller indices of this plane. (931)
5. Determine the number of the nearest neighbours and the closest distance of approach in terms of lattice parameter for monoatomic *sc*, *bcc* and *fcc* structures. (6, a; 8, $a\sqrt{3}/2$; 12, $a/\sqrt{2}$)
6. Calculate the linear density (number of atoms per unit length) along cube edge, face diagonal and body diagonal of an *fcc* unit cell of side length a. [$1/a$, $\sqrt{2}/a$, $1/(a\sqrt{3})$]
7. Nickel (*fcc*) has the lattice parameter of 3.52 Å. Calculate the atomic planar density (number of atoms per unit area) on (100), (110) and (111) planes. Is it possible to pack the atoms more closely than in (111) plane ? (1.61×10^{19}, 1.14×10^{19}, 1.86×10^{19} atoms m^{-2}; No)
8. Calculate the angles which [111] direction of a cubic lattice makes with [100] and [110] directions. (54°44', 35°15')
9. Show (111) and (222) planes in a cubic unit cell of side a. Compute the distances of these planes from a parallel plane passing through the origin. [$a/\sqrt{3}$, $a/(2\sqrt{3})$]
10. Calculate the distances between the adjacent parallel planes of the type (100), (110) and (111) in an *fcc* lattice of lattice constant a. Check the validity of the statement "The most close-packed planes are the most widely spaced." [$a/2$, $a/(2\sqrt{2})$ and $a/(\sqrt{3})$]
11. Copper (*fcc*) has density of 8960 kg m^{-3}. Calculate the unit cell dimension and the radius of Cu atom, given the atomic mass of Cu as 63.54 amu. (3.61 Å, 1.28 Å)
12. Prove that c/a ratio for an ideal *hcp* structure is 1.633.
13. Zinc (*hcp*) has lattice parameters a and c as 2.66 Å and 4.95 Å respectively. Calculate the packing fraction and density of zinc, given the atomic radius and the atomic mass of Zn as 1.31 Å and 65.37 amu respectively. (62%, 7155 kg m^{-3})
14. Calculate the distance between two atoms of a basis of the diamond structure, if the lattice constant of the structure is 5 Å. (2.17 Å)